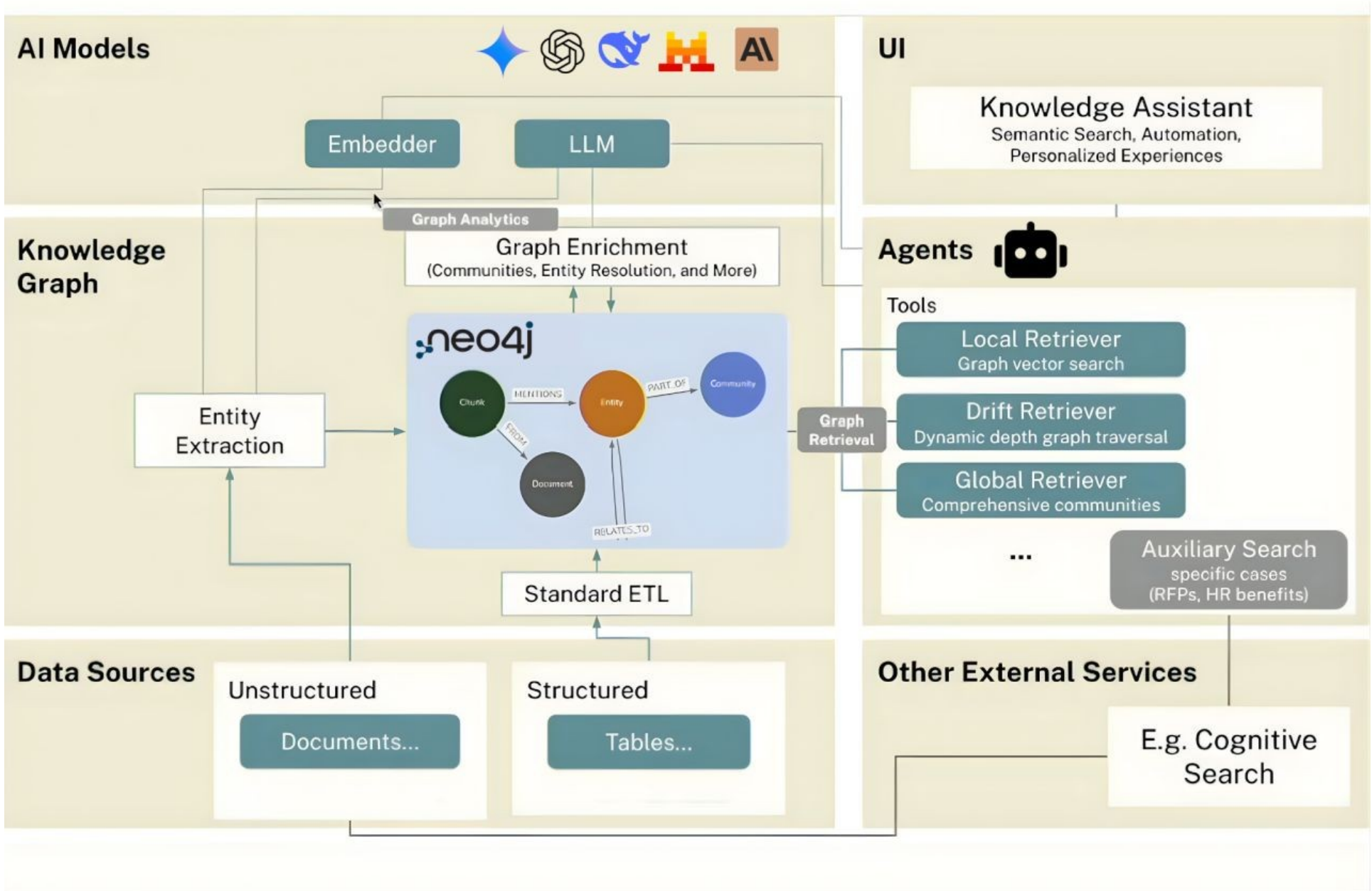
Agentic GraphRAG Architecture



Skills, skills, skills...



Module 1

Graph Basics

- **Creating a Graph from Structured Data**
- **Basic Cypher Queries and Pattern Matching**
- **Graph Algorithms**
- **Text Embeddings for Semantic Analysis**
- **Vector Search**

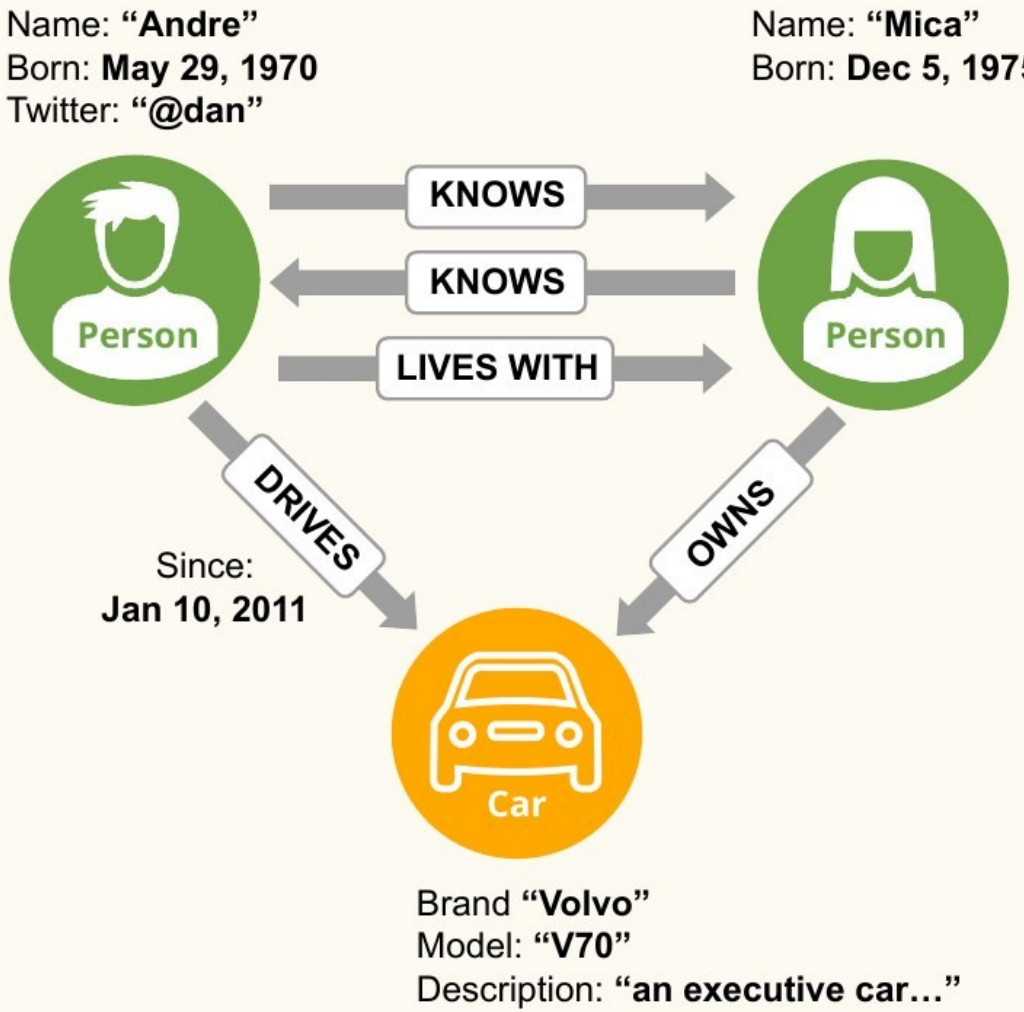
Knowledge Graph = design patterns to organize & access interrelated data

Property Graph Data Model

Nodes represent entities in the graph

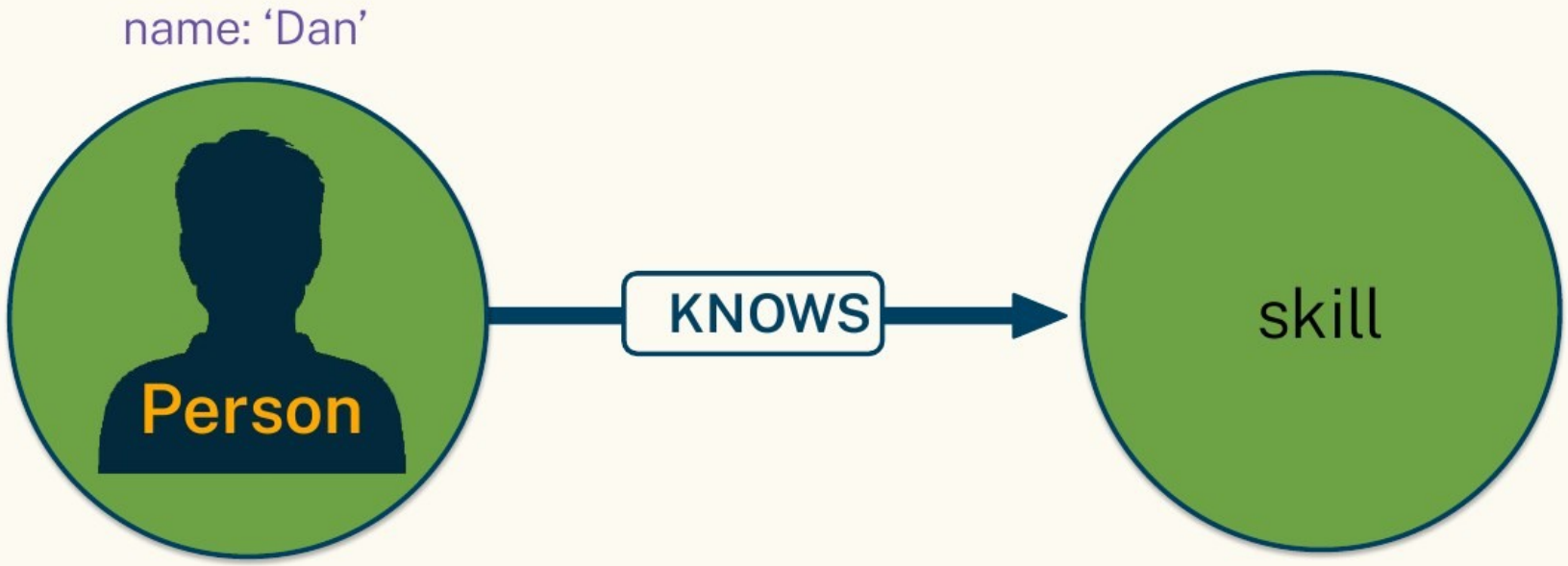
Relationships represent associations or interactions between nodes

Properties represent attributes of nodes or relationships



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Cypher: A Powerful & Expressive Query Language



NODE — RELATIONSHIP TYPE — NODE

```
MATCH (p:Person {name:"Dan"} ) -[:KNOWS]-> (skill)
```

VARIABLE — LABEL — PROPERTY — VARIABLE

```
RETURN p.name as person, skill
```


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50+ Graph Algorithms in Neo4j



Pathfinding & Search

- Shortest Path
- Single-Source Shortest Path
- All Pairs Shortest Path
- A* Shortest Path
- Yen's K Shortest Path
- Minimum Weight Spanning Tree
- K-Spanning Tree (MST)
- Random Walk
- Breadth & Depth First Search



Centrality / Importance

- Degree Centrality
- Closeness Centrality
- Harmonic Centrality
- Betweenness Centrality & Approx.
- PageRank
- Personalized PageRank
- ArticleRank
- Eigenvector Centrality



Community Detection

- Triangle Count
- Local Clustering Coefficient
- Connected Components (Union Find)
- Strongly Connected Components
- Label Propagation
- Louvain Modularity
- K-1 Coloring
- Modularity Optimization



Link Prediction

- Adamic Adar
- Common Neighbors
- Preferential Attachment
- Resource Allocations
- Same Community
- Total Neighbors



Similarity

- Euclidean Distance
- Cosine Similarity
- Node Similarity (Jaccard)
- Overlap Similarity
- Pearson Similarity
- Approximate KNN



Embeddings

- Node2Vec
- Random Projections
- GraphSAGE

... Auxiliary Functions:

- Random graph generation
- Graph export
- One hot encoding
- Distributions & metrics

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Today we will explore

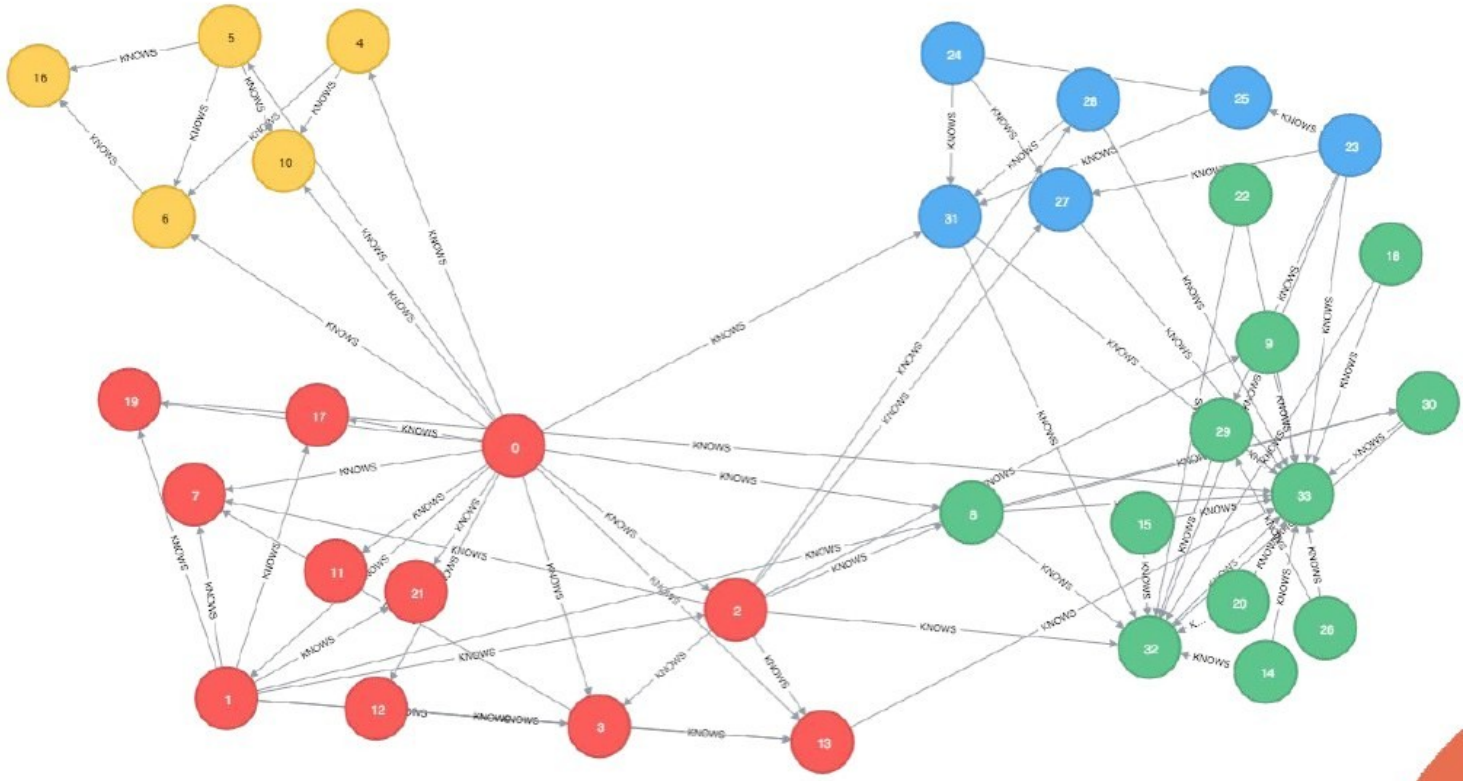
Community Detection

Evaluate how groups of nodes may be clustered or partitioned in a graph.

Community id properties assigned to node based on relationship structure.

Useful for:

- Segmentation
- Clustering
- Entity resolution
- Summarization (for AI)



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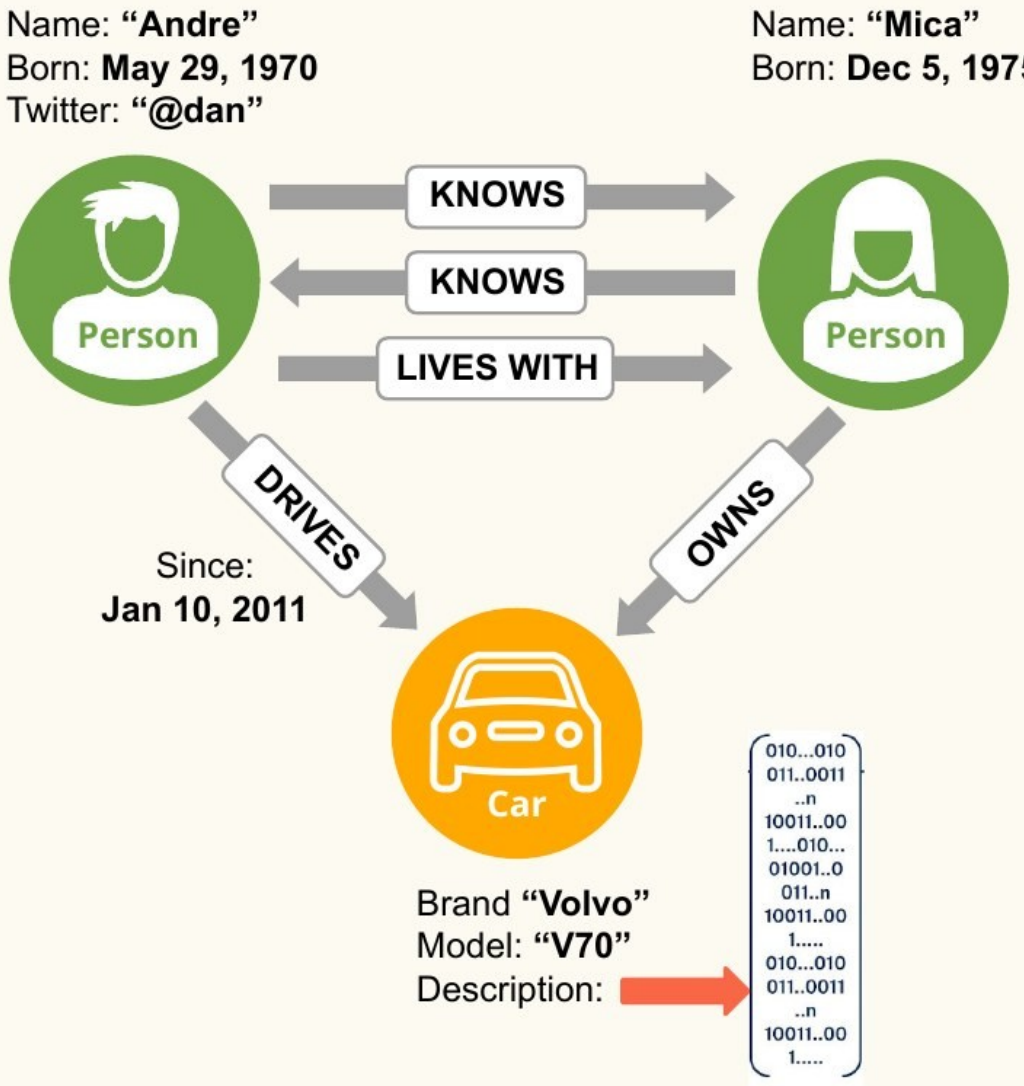
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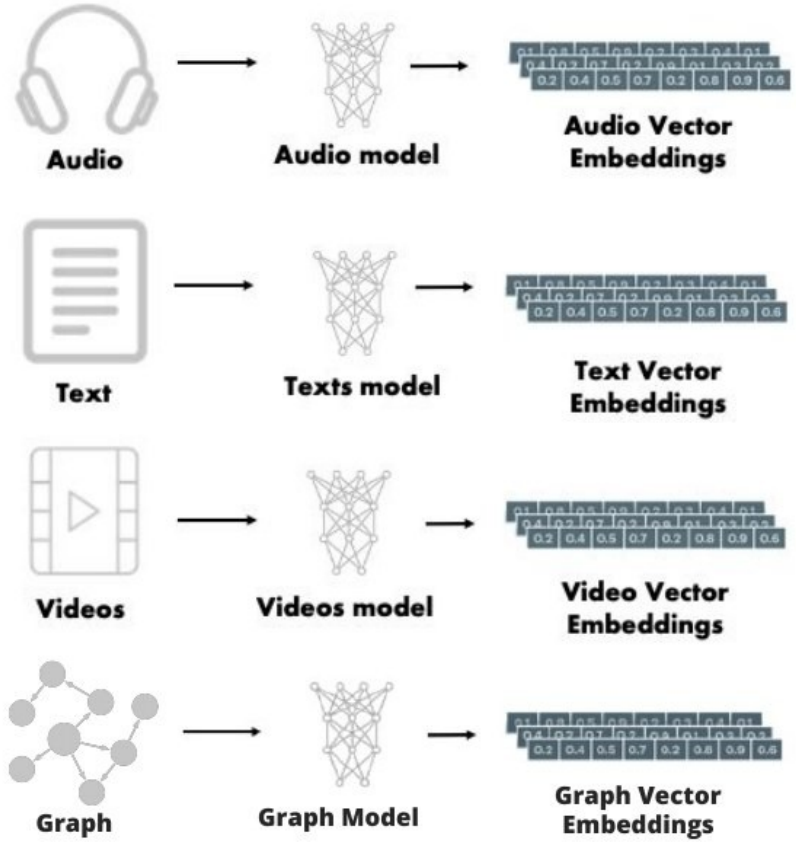
Properties represent attributes of nodes or relationships



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What are Embeddings?

- A type of data compression
- Transform messy data (text, images, audio, etc.) into a compact format for ML algorithms
- Most often numeric **vectors** (a.k.a arrays) with 100s or 1000s of elements
- Preserve information such that ***"similar" items have proportionally "similar" embedding vectors***
- Similarity is measured with vector algorithms (cosine, euclidean, etc.)



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Search & Vectors in Neo4j

Neo4j makes search efficient through a variety of indexes including

- **Range:** General index for predicates based on equality and range.
- **Point:** Predicts on geospatial points like distance bounding boxes, etc.
- **Text:** Predicates on strings like contains, ends with, etc.
- **FullText:** Text search based on tokenization and analyzers
- ***Vector: ANN (Approximate Nearest Neighbor) search on vectors***

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How is similarity measured

What would you use for openai embedding models?

For OpenAI embedding models, **cosine similarity** is the preferred choice. Here's why:

Why Cosine Similarity?

1. **OpenAI Embeddings Are Normalized**
   - OpenAI embedding models (like `text-embedding-ada-002`) generate **normalized vectors**, meaning each vector has a unit length (i.e., a magnitude of 1).
   - In such cases, **cosine similarity and Euclidean distance are proportional**, but cosine similarity is computationally more efficient.
2. **Focus on Semantic Meaning**
   - Since cosine similarity measures the **angle between vectors**, it captures semantic similarity between words, sentences, or documents effectively.
   - Euclidean distance, on the other hand, is sensitive to magnitude differences, which aren't meaningful in normalized embeddings.
3. **Industry Best Practice**
   - Most applications using OpenAI embeddings (e.g., search ranking, recommendation systems, and semantic clustering) rely on cosine similarity.
   - OpenAI's own documentation recommends using **cosine similarity or dot product** over Euclidean distance.

Example Pattern

Name: Graph Enhanced Vector Search

Description: The user question is embedded using the same embedder used to create chunk embeddings. A vector similarity search is executed on the chunk embeddings to find k (number previously configured by developer/user) most similar chunks. A traversal of the Domain Graph starting at the found chunks is executed to retrieve more context.

Context: The biggest problem with basic GraphRAG patterns is finding all relevant context necessary to answer a question. The context can be spread across many chunks not being found by the search. Relating the real-world entities from the chunks to each other and retrieving these relationships together with a vector search provides additional context about these entities that the chunks refer to. They can also be used to relate chunks to each other through the entity network.

Required pre-processing: Use an LLM to execute entity and relationship extraction on the chunks. Import the retrieved triples into the graph.

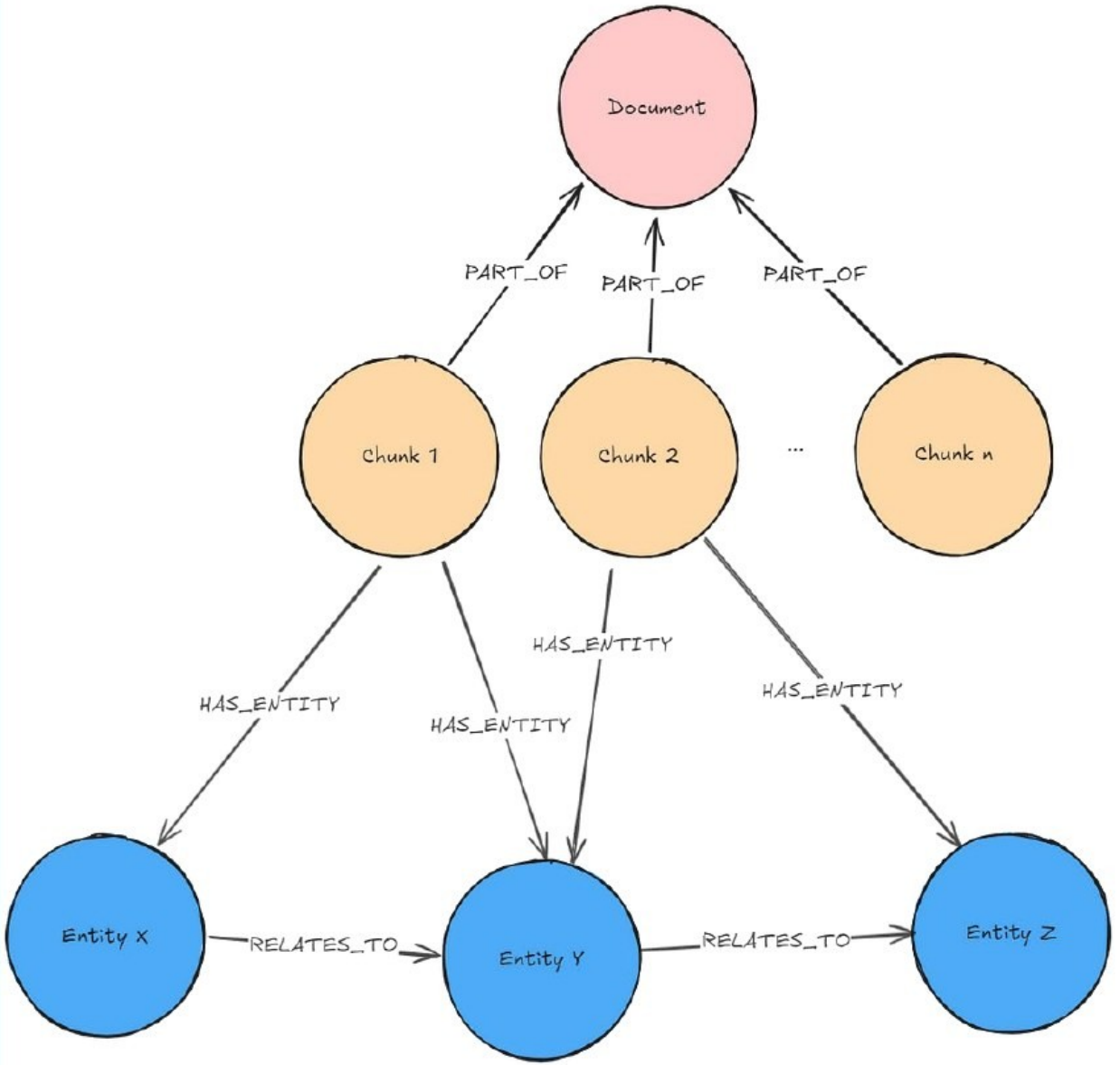
Variations: Entity disambiguation, Question-guided/Schema-defined extraction, Entity embeddings, Ontology-driven traversal

```
MATCH (node)-[:PART_OF]->(d:Document)
MATCH (node)-[:__HAS_ENTITY__]->(e)
MATCH path=(e)(()-[rels:!HAS_ENTITY&!PART_OF]-()){0,2}(:!Chunk&!Document)
...
RETURN ...
```

AKA: Graph + Vector, Augmented Vector Search

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Required graph pattern: Lexical Graph with Extracted Entities



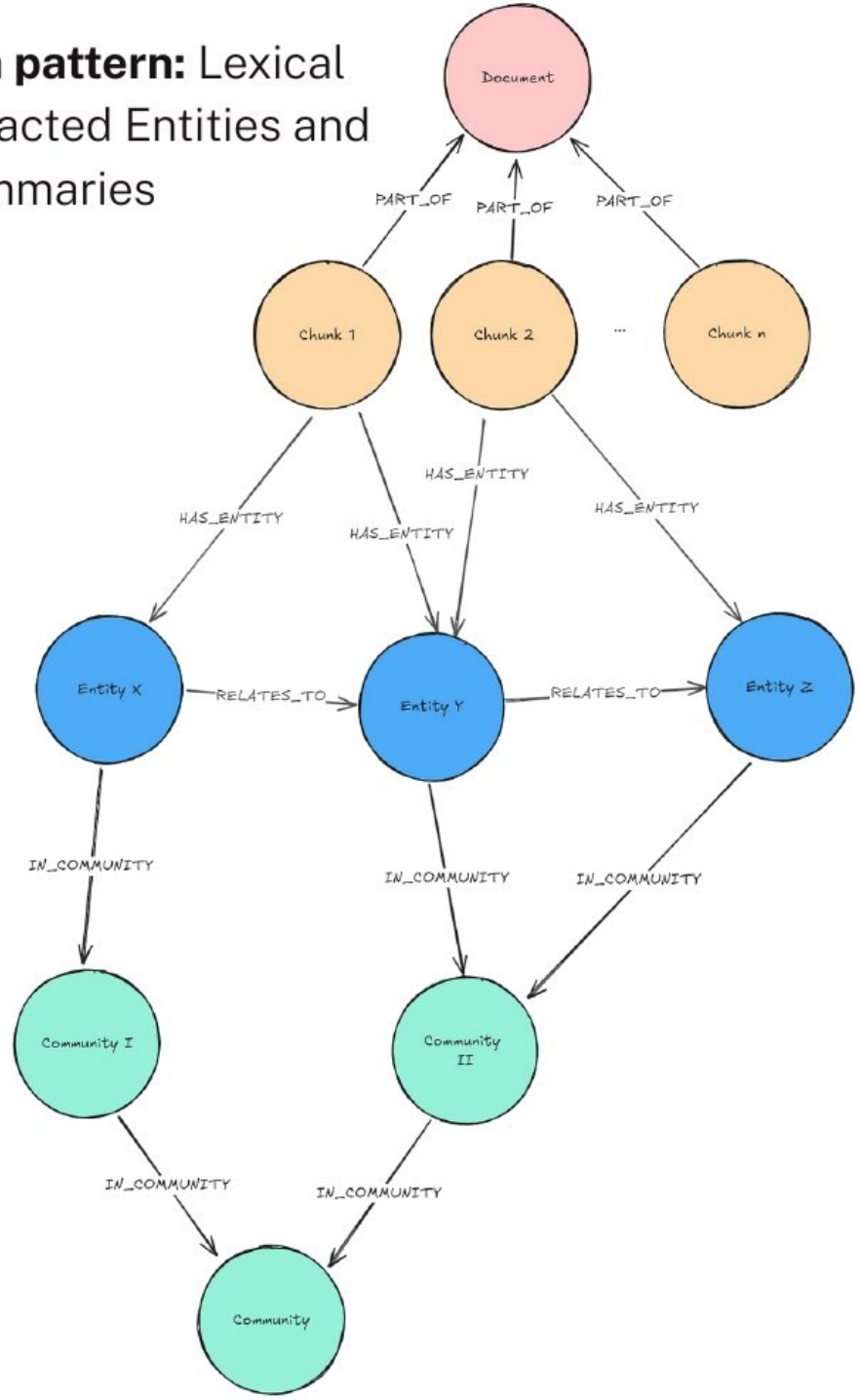
Graph Enhanced Vector Search

Global Community Summary Retriever

AKA: *Microsoft GraphRAG*, Global Retriever
Context: Certain ***questions that can be asked on a whole dataset*** do not just relate to things present in some chunks but rather search for an overall message that is overarching in the dataset.
Required pre-processing: In addition to extracting entities and their relationships, we need to form hierarchical communities within the Domain Graph. For every community, an LLM summarizes the entity and relationship information into Community Summaries.

```
MATCH (c:__Community__)
WHERE c.level = $level
RETURN c.full_content AS output
```

Required graph pattern: Lexical Graph with Extracted Entities and Community Summaries



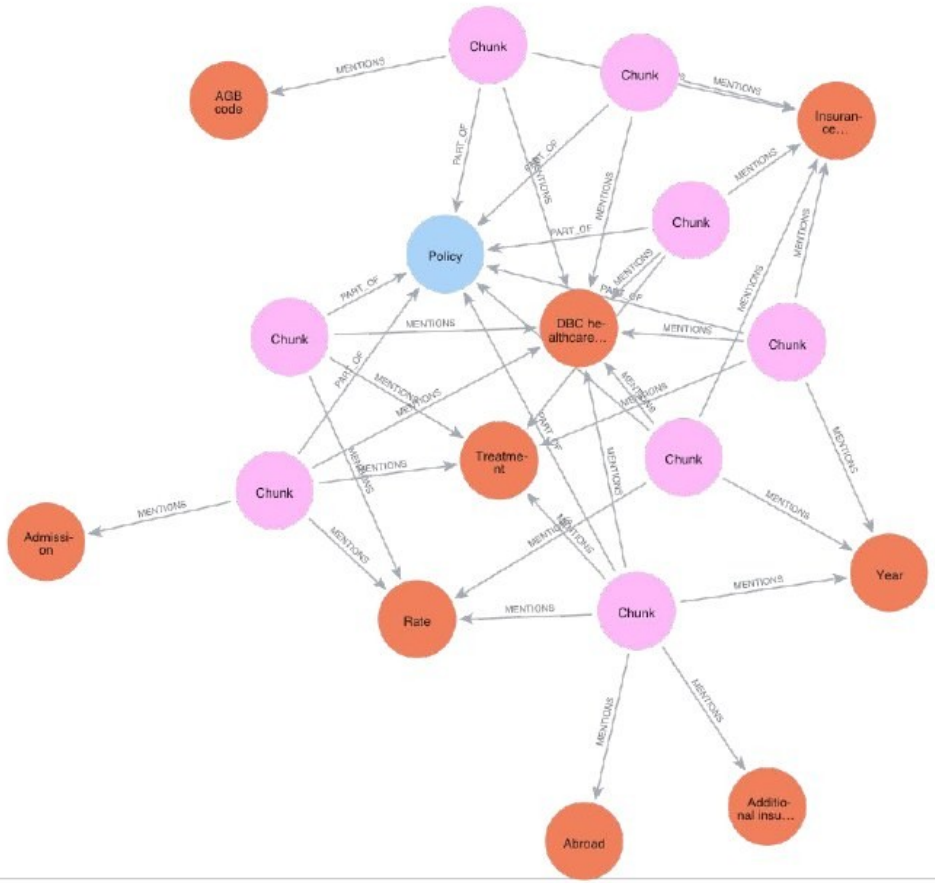
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Add Domain Knowledge

- There are references in the document that give more context
- These add more context to the chunks

See also:

- Abroad (definitions)
- DBC healthcare product (definitions)
- Rate (definitions)
- Treatment (definitions)
- Year (definitions)




```
1 MATCH (c:Chunk)-[:MENTIONS]-(d:Definition)
2 WHERE c.id in $chunk_ids
3 WITH DISTINCT d as d
4 RETURN d.definition as definition, d.description as description
```

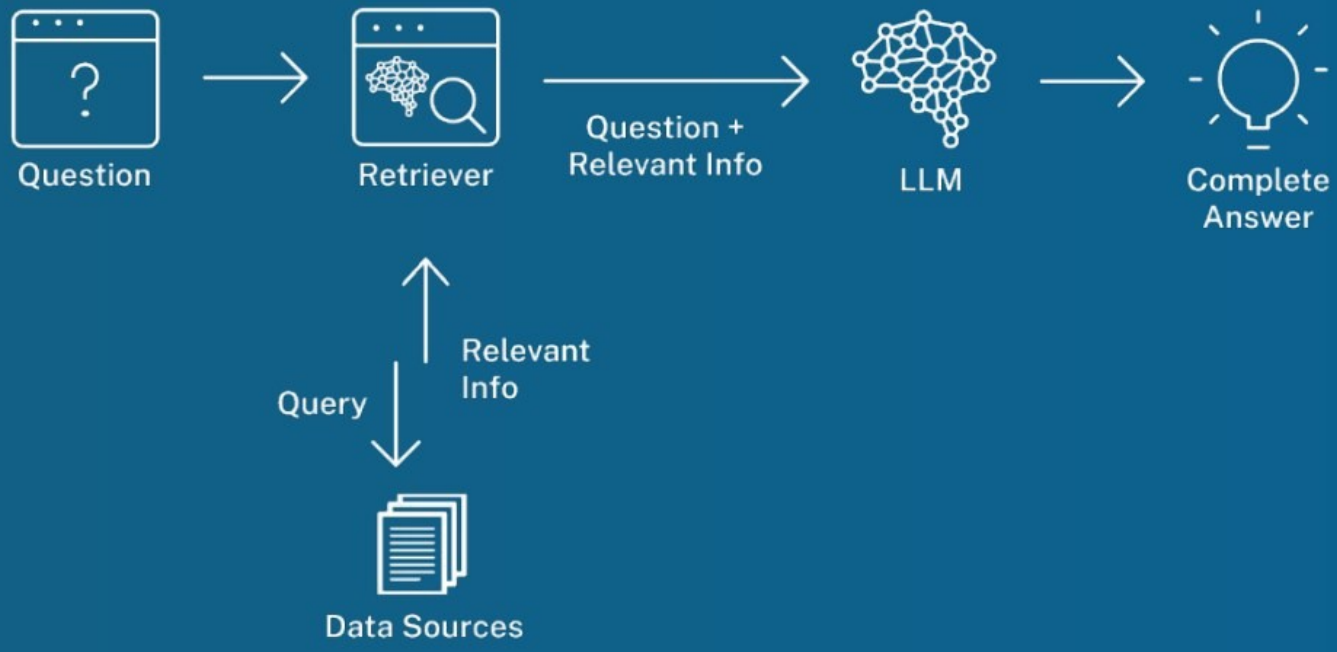
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Module 3

GraphRAG & Agents

- **Experiment with queries for an Agent**
- **Define Tooling**
- **Create an agents with the available tools**
- **Chatbot for an Agent**
- **Text2Cypher (if we got time)**

Ordinary RAG



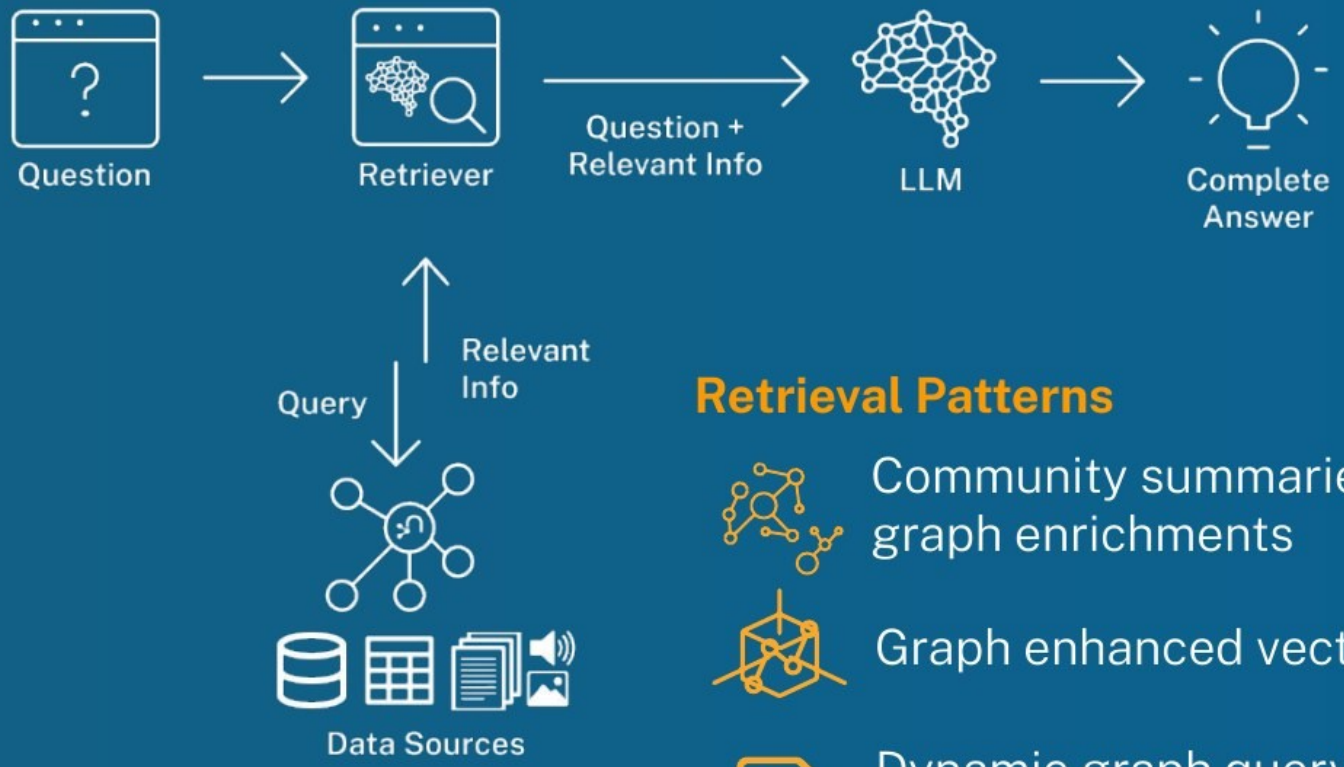
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GraphRAG: RAG with graph

It's actually quite broad…

GraphRAG is Retrieval Augmented Generation (RAG) using Knowledge Graphs

Improves GenAI by taking advantage of rich graph data structures

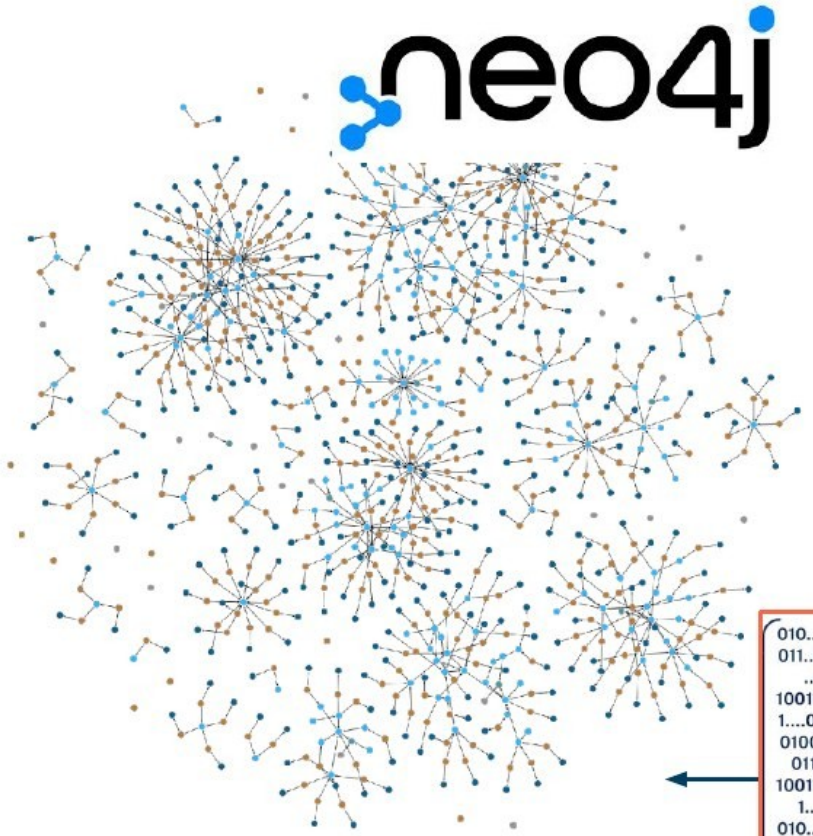


Retrieval Patterns

- Community summaries & graph enrichments
- Graph enhanced vector search
- Dynamic graph query generation
- + more: graph vectors, parent-child retrievers, etc.

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Accurate, Contextual and Explainable



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Vector Embedding

Embedding Model

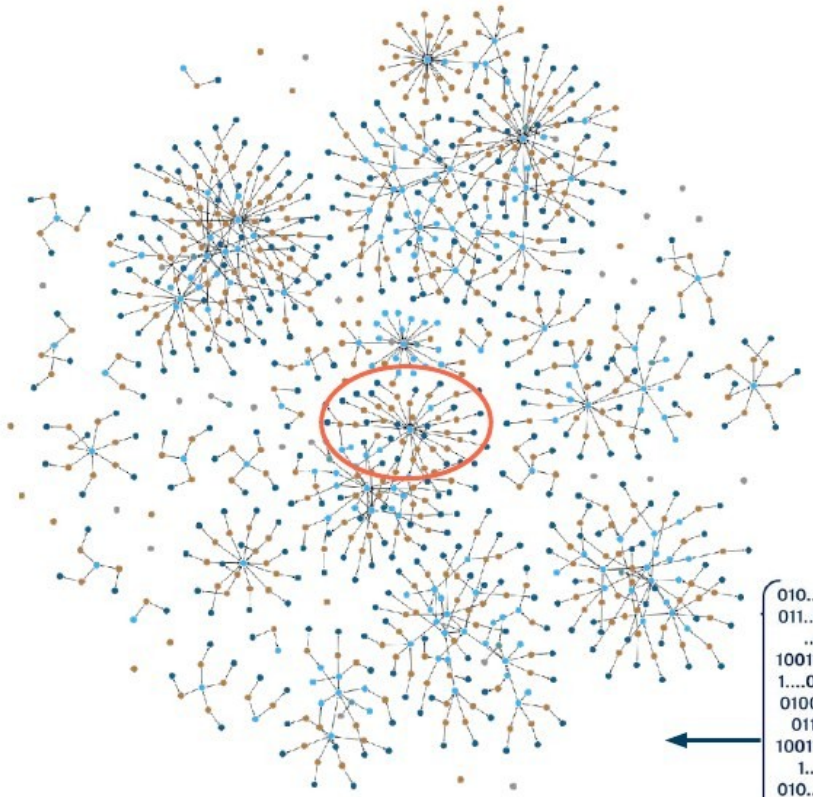
User Question

GenAI App

Who is the main respondent of the AWS RFP?

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Accurate, Contextual and Explainable



Similarity Search using
Neo4j Vector Index

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Vector Embedding

Embedding Model

User Question

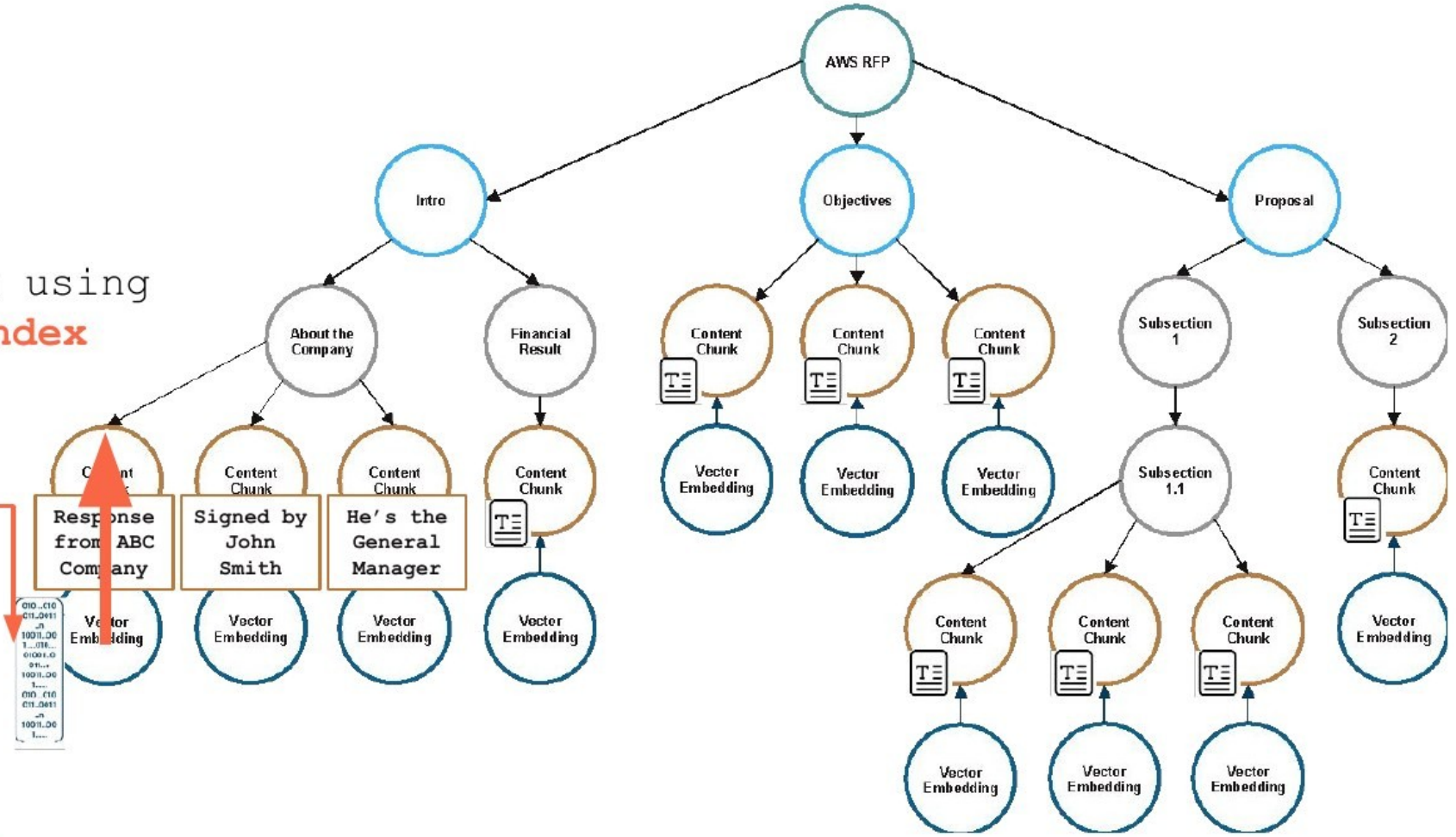
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Accurate, Contextual and Explainable

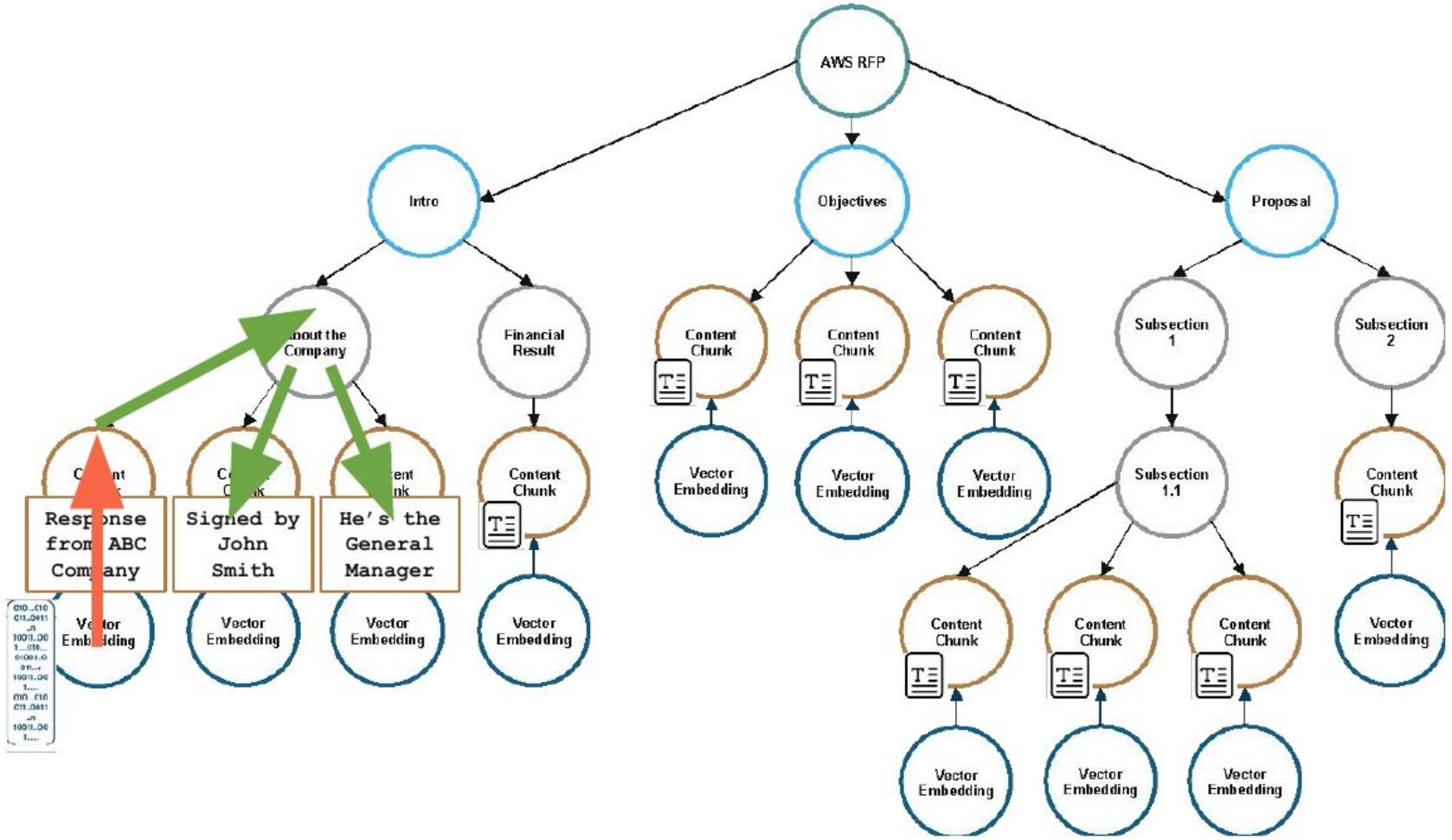
Similarity Search using
Neo4j Vector Index



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Accurate, Contextual and Explainable

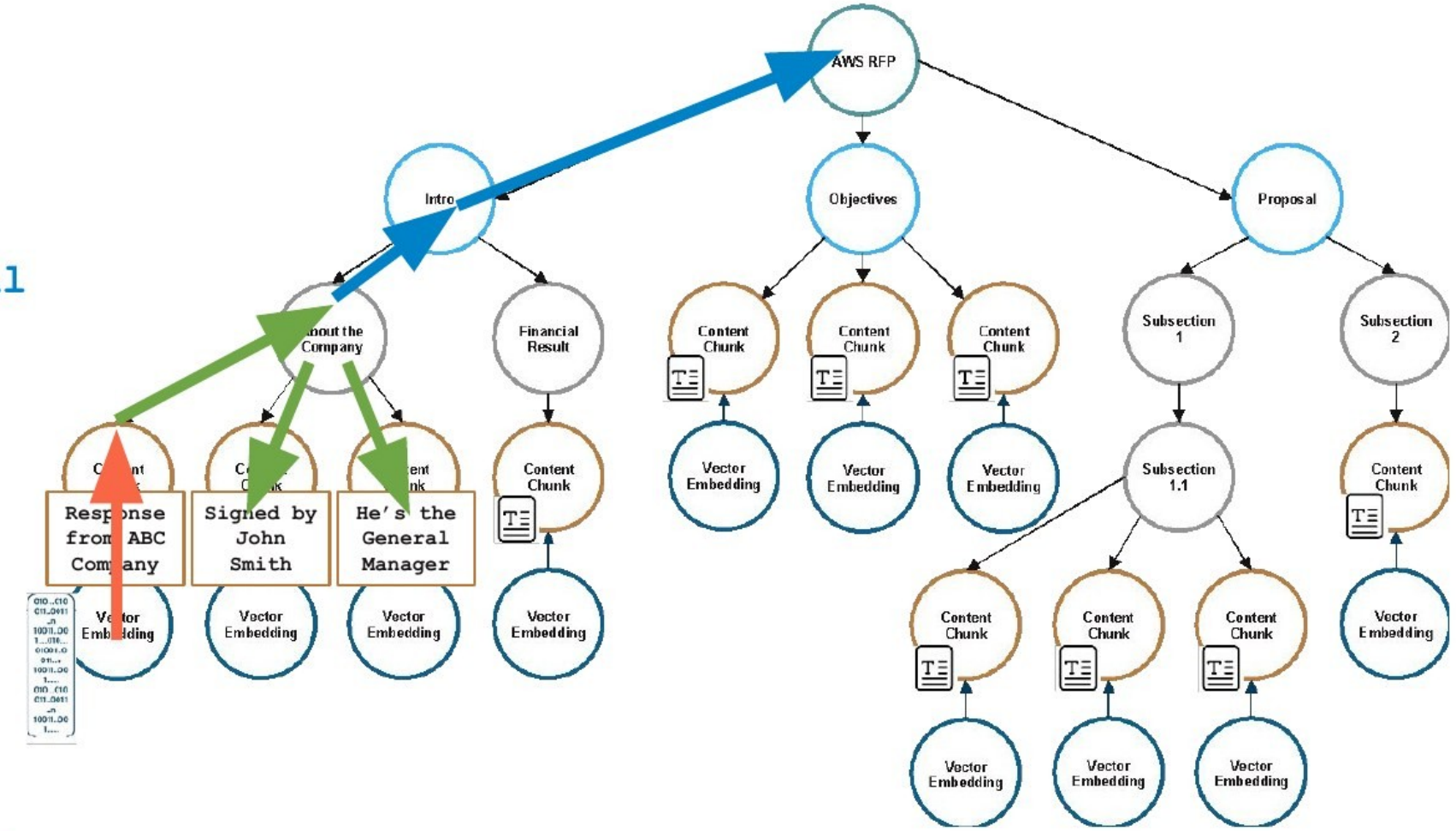
Contextual Knowledge Retrieval within Neo4j KG



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Accurate, Contextual and Explainable

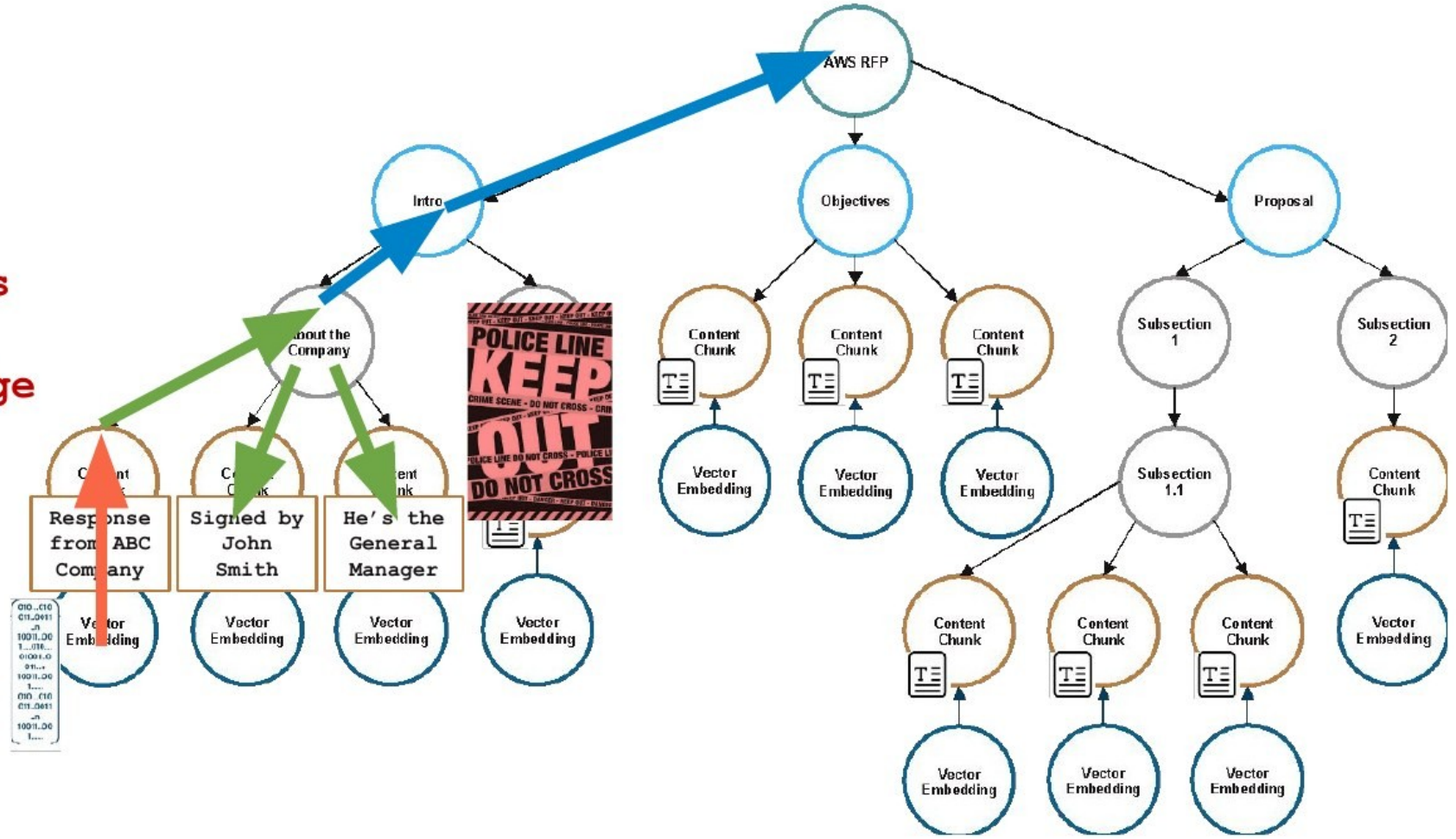
Knowledge Retrieval
to aid in
Explainability



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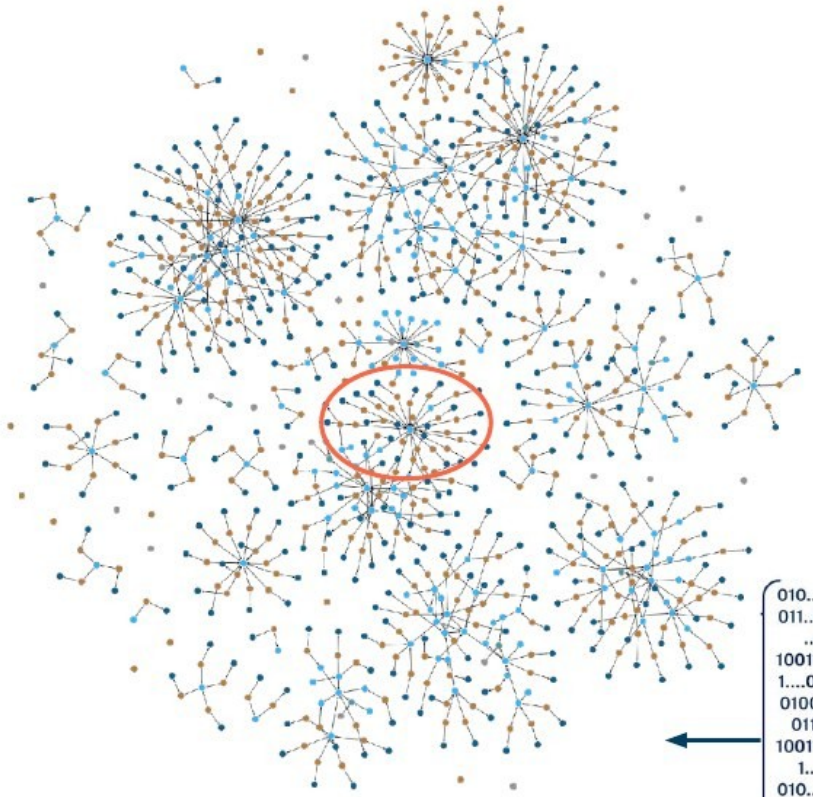
Accurate, Contextual and Explainable

Fine Grained Access Control to prevent **unwarranted Knowledge Retrieval**



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Accurate, Contextual and Explainable



Similarity Search using
Neo4j Vector Index

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Vector Embedding

Embedding Model

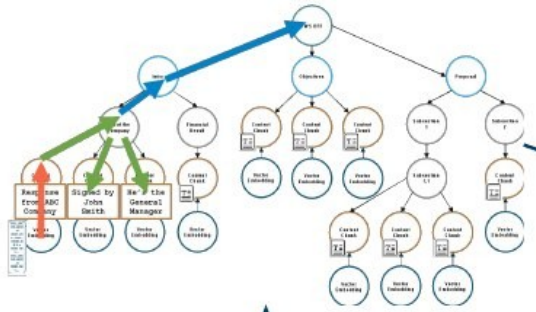
User Question

GenAI App

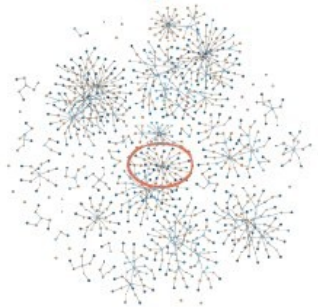
Who is the main respondent of the AWS RFP?

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Accurate, Contextual and Explainable



Similarity Result



Similarity Search using Neo4j Vector Index

010...010
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1....010...
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011..n
10011..00
1.....
010...010
011..0011
..n
10011..00
1.....

Vector Embedding

Bedrock LLM

Embedding Model

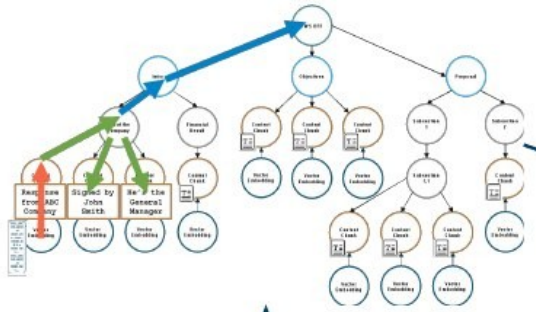
User Question

GenAI App

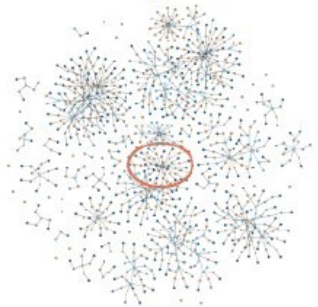
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Accurate, Contextual and Explainable



Similarity Result

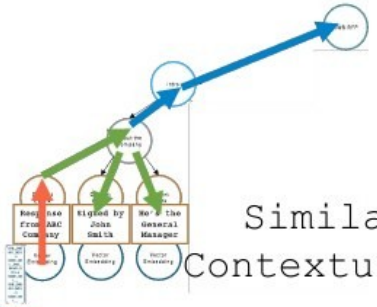


Similarity Search using Neo4j Vector Index

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Vector Embedding

Embedding Model



Similarity + Contextual Result

Bedrock LLM

The main respondent of the AWS RFP is John Smith, General Manager XYZ Group representing ABC Company (source: RFP_AWS.pdf, page: 38)

User Question

GenAI App

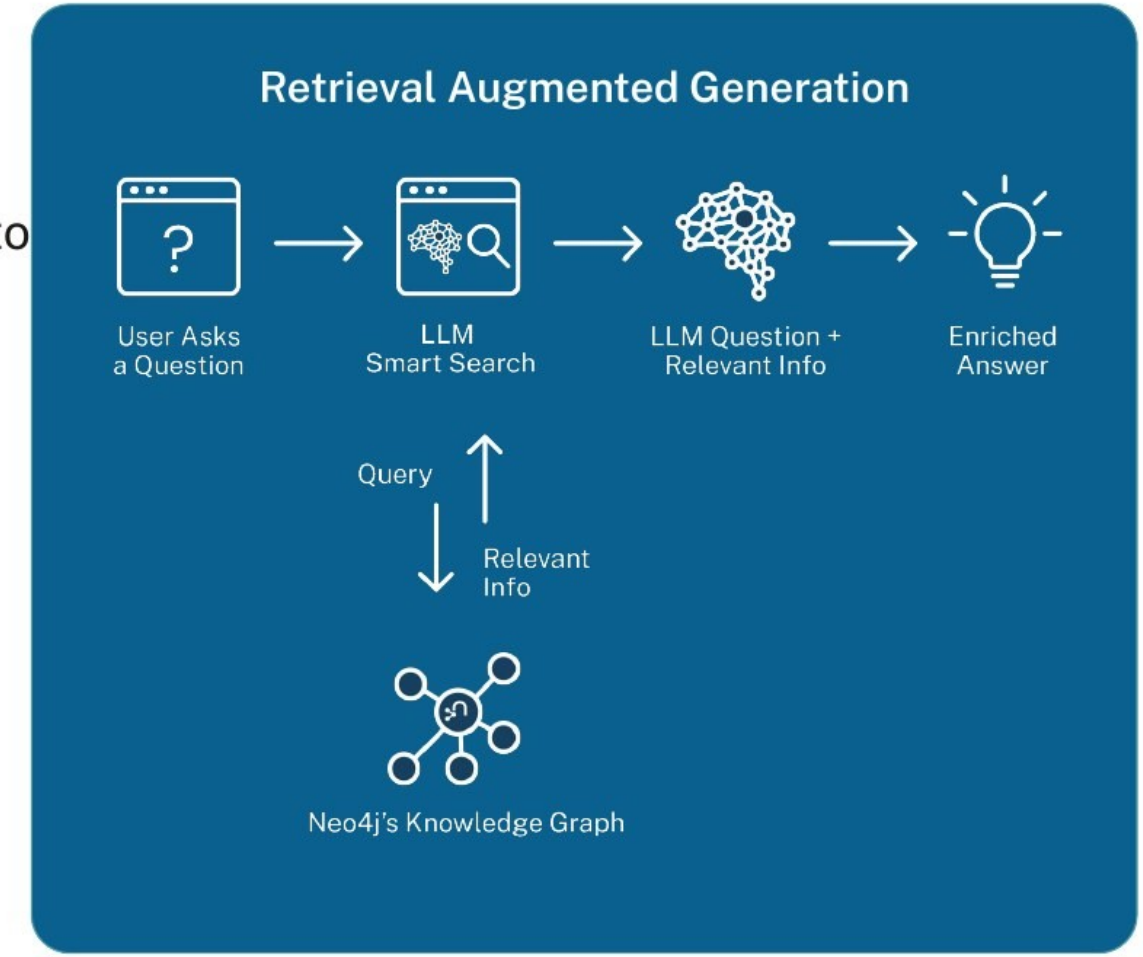
Who is the main respondent of the AWS RFP?

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Neo4j Knowledge Graph as Database of Truth

A Neo4j Knowledge Graph combined with LLM's obtain improvements:

- **Accuracy** - Obtain better answers compared to plain vector searches
- **Explainability**: Provide the user with more reasoning on how the results were obtained.
- **Acceleration**: Having all the capabilities in one platform increases understanding and decreases time to value.



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